

Question 6

PAC Problem Analysis Chart

Input

Vehicle type

Battery

Required charging

Parking duration

~~membership~~ ~~membership~~

membership

Station availability

Required Result

Charging status

Charging cost

Charging priority

Parking cost

Discount

Final payable

Long-stay warn.

Final details

Process Required

Check vehicle type

and charging eligibility

Calculate charging

need

Calculate charging

cost

Calculate parking

charges

Check peak/off-peak

hours

Apply membership discount

Apply priority/priority

Check charging

availability.

Solution Alternative

use of if-else

use of switch.

use of for loop.

Date: _____

IPO Chart

Input	Process	Output
vehicle type	Check vehicle eligibility	charging status
Battery	Calculate charging needed	charging priority
Rejected %	Calculate charging cost	Peak / off status
Parking duration	Calculate parking cost	Charging cost
Current time	Apply discounts	Parking Cost
membership	Apply priority / free parking	Discount
Disability	Calculate total	Final amount
Station availability		long stay / standard message

Algorithm

1. Start
2. Take vehicle, shelter, parking, time, membership and priority as inputs.
3. Checks for the charging station availability and vehicle eligibility.
4. Apply charging priority, peak ~~on~~ / peak off rates and discounts.
5. Calculate parking charges according to parking discount and rates.
6. Calculate final bill ~~to~~ to be paid.
7. Display the charges and long say message if applicable.
- ~~8. Stop.~~
8. Stop.

Pseudo code

Begin

If Input vehicle type, battery, required charge, duration, time, membership, disability, and station availability

Set charging cost, parking cost and discounted

If charging station is not available

Display "charging unavailable"

else

Check vehicle eligibility

Calculate charging needed

end if-else

If charging required

Determine charging priority

Determine peak / peak-off charging time

Calculate charging cost and applicable discount

end If

Calculate parking cost based on duration

If disabled priority available by

Set parking cost to 0

end if

else if member is ~~data~~ confirmed

Apply 20% discount

end else if

Calculate final amount = charging - cost + p-cost

If duration > 8

Display long stay warning

else

Display standard parking message

end if - else

Display final details.

End.

start

char type, member, disabled,
station

int, battery, required, duration,
time, charging needed,
float charging-cost, parking cost,
discount, final Amount.

input vehicle type, battery,
required, duration, time,
membership, disability,
charging station availability

No

if
station availability = Y

if
vehicle type =
H

Yes

Output
charging unavailable
- Parking only

Output
No charger available

A

if
vehicle type =
H

No

Yes

Battery ≥ 40

Yes

Output vehicle not
quality of ~~to~~ to charge

else if
vehicle type
E

Output : vehicle qualify to
charge

B

B

charging needed = required - battery

if
required =
battery

yes

Output no charge is
required

No

if
battery < 5
or required
≥ 80

No

if
disabled = Y or
member = F
and
battery < 30

No

else

yes

yes

yes

if
peak = Y

if
peak = Y

if
peak = Y

~~charging cost 50~~
discount = 0

~~charging cost 50~~
discount =
100 - pet %

discount
= 0

discount =
member
pet %

discount = 0

discount =
member
pet %

charging cost 50

charging cost 50

charging cost 50

C

